Q.1 Which of the following is true about the existence of free variables (parameter) in a system of Linear Equations?

A. They guarantee the Inconsistency.

B. None of above all

C. They do not guarantee the Consistency.

D. They guarantee the Consistency.

Q.2 $$\begin{gathered} The ~values~ of~ x~ and~ y~ which~ satisfy~ the ~matrix~ equation: \ \left({\begin{array}{*{20}{c}} 4 \\ y \end{array}} \right) = x\left({\begin{array}{*{20}{c}} 2 \\ 3 \end{array}} \right) are - - - - . \\ \end{gathered}$$

A. $$({{array}{*{20}{c}} x \\ y {array}}) = ({{array}{*{20}{c}} 3 \\ 2 {array}})$$

B. $$({{array}{*{20}{c}} x \\ y {array}}) = ({{array}{*{20}{c}} 2 \\ 6 {array}})$$

C. $$({{array}{*{20}{c}} x \\ y {array}}) = ({{array}{*{20}{c}} 2 \\ 3 {array}})$$

D. $$({{array}{*{20}{c}} x \\ y {array}}) = ({{array}{*{20}{c}} 6 \\ 2 {array}})$$

Q.3 If a matrix is in reduced row echelon form, then it is also in row echelon form.

A. False

B. None of the above

C. May be

D. True

Q.4 $$\begin{gathered} {\text{Which of the following is corresponding Matrix form of }} \ {\text{the Linear equation }}x + y = 3? \\ \end{gathered}$$

A. $$x({{array}{*{20}{c}} 1 \\ 0 {array}}) + y({{array}{*{20}{c}} 0 \\ 1 {array}}) = 3({{array}{*{20}{c}} 0 \\ 1 {array}})$$

B. $$x({{array}{*{20}{c}} 1 \\ 0 {array}}) + y({{array}{*{20}{c}} 0 \\ 1 {array}}) = 3({{array}{*{20}{c}} 1 \\ 0 {array}})$$

C. $$x[1] + y[1] = 3[1]$$

D. Cant be expressed in Matrix form.

Q.5 $${\text{Which of the following would be the value of t, if}} \left[{\begin{array}{*{20}{c}} {2{e^{ - 3t}}} \\ {5{e^{ - 3t}}} \end{array}} \right] = \left[{\begin{array}{*{20}{c}} 2 \\ 5 \end{array}} \right]?$$

A. $${{[{{array}{*{20}{c}} 2 \\ 5 {array}}]}}{{[{{array}{*{20}{c}} 2 \\ 5 {array}}]}} = 1$$

B. $$ {1}{3}$$

C. zero

D. $$[{{array}{*{20}{c}} 1 \\ 1 {array}}]$$

Q.6 $$\begin{gathered} {\text{For the}} ~{\text{Linear Equation}}: {x_1} + {x_2} + {x_3} + {x_4} = 1,{\text{how many Parametres}} ~{t_{\text{i}}}{\text{'s }} \ {\text{would have to introduce in order to find its solution?}} \\ \end{gathered}$$

A. 3

B. 2

C. 1

D. 4

Q.7 $$\begin{array}{*{20}{c}} {{\text{For the following system of Equations:}}} \\ {x + 2y = 0,{\text{ }}2x - y = 0{\text{;}}} \\ {{\text{the solution set is - - - - - }}{\text{.}}} \end{array}$$

A. $${{Real line}}~ {R}$$

B. $${{Line through origin in}} ~{{R}^3}$$

C. $${{Line through origin in}} ~{{R}^2}$$

D. $${{Any arbitrary line in}} ~{{R}^2}$$

Q.8 $If{\text{ }}A = \,\left({\begin{array}{*{20}{c}} a&b \\ c&d \end{array}} \right),{\text{ }}then{\text{ }}which{\text{ }}of{\text{ }}the{\text{ }}following{\text{ }}is\,\,\left| A \right|$

A. bc+ad

B. ad-bc

C. ad+bc

D. bc-ad

Q.9 $$\begin{gathered} {\text{Under which of the following condition, a system of Linear Equations whose }} \ {\text{Row - Reduce form is }}\left({\begin{array}{*{20}{c}} {\begin{array}{*{20}{c}} 1 \\ 0 \end{array}}&{\begin{array}{*{20}{c}} 2 \\ 0 \end{array}}&{\begin{array}{*{20}{c}} | \\ | \end{array}}&{\begin{array}{*{20}{c}} { - 1} \\ {h - 3k} \end{array}} \end{array}} \right){\text{ has }}~Infinite ~{\text{many}} ~{\text{solutions?}} \\ \end{gathered}$$

A. $$h 3k$$

B. $$({h,k}) ({0,0})$$

C. $$h = 3k$$

D. $$({h,k}) = ({0,0})$$

Q.10 $$How ~many ~Pivot ~positions ~the ~matrix:\left({\begin{array}{*{20}{c}} {\begin{array}{*{20}{c}} 2&3&1 \end{array}} \\ {\begin{array}{*{20}{c}} 4&6&2 \end{array}} \end{array}} \right) will~ have?$$

A. 4

B. 1

C. 2

D. 3
